

Imran E Kibria

4th YEAR PHD RESEARCHER IN SPEECH & AUDIO AI

[Website](#) [Github](#) [LinkedIn](#) [Google Scholar](#) [✉ kibria.5@osu.edu](mailto:kibria.5@osu.edu)

Research Interests

Speech Quality Assessment, Speech Enhancement, Domain Generalization, Audio Foundation Models.

Education

Ohio State University

Ph.D. in Computer Science & Engineering, GPA: 3.9 / 4.0

Columbus, OH, USA

Aug 2023 - Dec 2027

- Research: Neural Representations for Perceptual Speech Quality
- Advisor: [Donald S. Williamson](#)

National University of Sciences and Technology

BS in Electrical Engineering

Islamabad, Pakistan

Sept 2016 - July 2020

- Major: Signal Processing and Machine Learning
- Thesis: Real-time Adaptive Filtering for Robust Vibration Sensing

Experience

Graduate Teaching Associate

Department of Computer Science & Engineering, Ohio State University

Columbus, OH, USA

Aug 2023 - Present

Courses

- Software I: Software Components — Instructor (Su'25, Au'25, Au'26)
- Machine Learning and Statistical Pattern Recognition (Sp'26)
- Computer Programming in Python (Au'23, Sp'24, Sp'25)
- Introduction to Neural Networks (Au'24)

Graduate Research Associate

Audio, Speech, & Perceptually-Inspired Research Lab, Ohio State University

Columbus, OH, USA

May 2023 - Aug 2023

Design Engineer

Center for Advanced Research in Engineering

Islamabad, Pakistan

Nov 2020 - May 2022

Developer Intern

N-ovative Health Technologies

Islamabad, Pakistan

July 2020 - Oct 2020

Publications

Sharpness-Aware Minimization for Generalizable Speech Quality Assessment

[In Review]

Imran E Kibria, Donald S. Williamson

IEEE/ACM TASLP

A Generalization Strategy for Speech Quality Prediction: Domain-Specific to Unified Datasets

May 2026

Imran E Kibria, Ada Lamba, Donald S. Williamson

IEEE ICASSP 2026

AttentiveMOS: A Lightweight Attention-Only Model for Speech Quality Prediction

Aug 2025

Imran E Kibria, Donald S. Williamson

ISCA Interspeech 2025

Smartphone-Based Point-of-Care Urinalysis Assessment

July 2022

Imran E Kibria, Hussnain Ali, Shoab A. Khan

IEEE EMBC 2022

Select Projects

One Model for All Domains: Zero-Shot Speech Quality Assessment [\[Code\]](#)

- Analyzed factors influencing domain shift in MOS-label datasets.
- Improved generalization by optimizing model weights to a flat minima.
- Investigated loss landscape characteristics and quality of latent features.
- Reported improvement over AdamW optimizer on 8 of 12 evaluation sets.

Skills: Domain Generalization, Sharpness-Aware Minimization, TensorFlow, Pytorch, Python.

Lightweight AI for Perceptual Speech Quality on the Edge [\[Code\]](#)

- Designed MOS prediction network for edge use case using 86K params.
- Analyzed transformer variants & positional encodings as design choices.
- Countered subjective noise in MOS ratings using self-knowledge distillation.
- Wav2vec-level performance with 100× fewer parameters on SOMOS dataset.

Skills: Swin Transformers, Self-Knowledge Distillation, TensorFlow, Python, SHEET toolkit.

Self-Supervised Learning for Speech Quality Assessment [\[Code\]](#)

- Adapted AST, CLAP, HuBERT, wav2vec 2.0, and WavLM for MOS prediction.
- Evaluated transfer learning performance on the NISQA benchmark.
- Compared SSL representations for robustness and cross-dataset generalization.

Skills: Foundation Models, Transfer Learning, Domain Generalization, PyTorch, HuggingFace.

Smartphone-Based Urine Test Strip Analysis for At-Home Diagnostics [\[Code\]](#)

- Developed algorithm for automated urine test strip analysis using smartphone camera.
- Performed lighting & color calibration, then color matching in different colors spaces.
- Achieved 92% in-house accuracy; low-cost alternative to laboratory analyzers.

Skills: Image Preprocessing & Segmentation, Multivariate Regression, Color Spaces, Scikit-learn.

Technical Skills

Languages: Python, MATLAB, Java, C, C++, SQL, LaTeX.

Deep Learning: Tensorflow, PyTorch, HuggingFace, Github

Speech Toolkits: SHEET, ESPNet, SpeechBrain

Awards & Honors

CCBS Summer Graduate Research Award	June 2026
Summer grant awarded by The Center for Cognitive and Brain Sciences (CCBS)	
CSE Scarlet & Gray Award	Aug 2023
Merit-Based Scholarship, The Ohio State University (5-year award)	
Best English Speaker	June 2017
SEECs (School of Electrical Engineering & Computer Science) Declamation Contest	
Certificate of Excellence	Dec 2014
Third Place – Secondary Board Examination, Cadet College Hasanabdal	